

Lecture 5:

Model Selection, Evaluation, and Assessment

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Plan for Today

Selection & Evaluation

Information Criteria

Cross-Validation

Bias-Variance Trade-off

Model Selection vs. Evaluation

Model Selection

The process of choosing the best model (or hyperparameters) from a set of candidates.

- ▶ **Goal:** Select the model structure.
- ▶ **Tool:** Validation sets, Information Criteria.

In sample fit of the data

The Pitfall

Overusing a single validation set for selection can lead to overfitting that specific subset of data.

Model Evaluation

Measuring how well the *final* model performs on unseen data.

- ▶ **Goal:** Estimate real-world error.
- ▶ **Tool:** Test sets, Cross-Validation.

The Confusion Matrix

A confusion matrix summarizes a classification models performance by comparing predicted vs. actual values.

- ▶ **TP**: Correctly identified positive.
- ▶ **TN**: Correctly identified negative.
- ▶ **FP (Type I)**: False alarm.
- ▶ **FN (Type II)**: Missed detection. *say 3 no consistent there is*

		Predicted	
		Pos	Neg
Actual	Pos	TP	FN
	Neg	FP	TN

this just shows the count of the different groups

Common Metrics for Model Evaluation

- ▶ **Accuracy:** Fraction of correct predictions. (*Caution: Misleading in imbalanced datasets*).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- ▶ **Precision:** Purity of positive predictions. Vital when **False Positives** are costly (e.g., spam filters).

$$\text{Precision} = \frac{TP}{TP + FP}$$

- ▶ **Recall (Sensitivity):** Ability to find positive instances. Vital when **False Negatives** are dangerous (e.g., cancer detection).

$$\text{Recall} = \frac{TP}{TP + FN}$$

- ▶ **F1-Score:** The harmonic mean of Precision and Recall. Useful when you need a balance between the two.
- ▶ **ROC & AUC:** Plots TPR vs. FPR. AUC (Area Under Curve) summarizes performance across *all* thresholds.

Introduction to Information Criteria

Goal: Select a model that explains the data well without being overly complex.

Information criteria penalize the likelihood based on the number of parameters (k).

Akaike Information Criterion (AIC)

$$AIC = 2k - 2\ln(L)$$

- ▶ Estimates out-of-sample prediction error.
- ▶ Penalizes complexity ($2k$).

Bayesian Information Criterion (BIC)

$$BIC = \ln(n)k - 2\ln(L)$$

▼ In sample Selection

- ▶ penalizes complexity more heavily when n is large.
- ▶ Tends to favor simpler models than AIC.

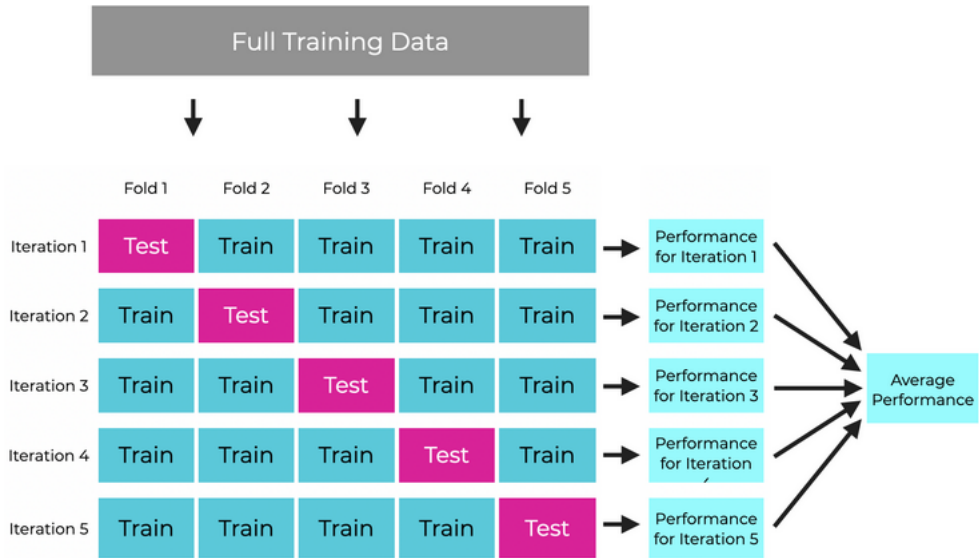
Rule of thumb: Lower values are better. Compare only on the same dataset.



The Procedure:

1. Shuffle the dataset.
2. Split into k groups (folds).
3. For each unique group:
 - ▶ Take the group as a hold-out or test data set.
 - ▶ Take the remaining groups as a training data set.
 - ▶ Fit a model on the training set and evaluate it on the test set.
4. Summarize the skill of the model using the sample of model evaluation scores.

k -Fold Cross-Validation



Stratified k -fold

Ensures each fold maintains the same class distribution as the full dataset. **Essential for imbalanced classes.**

Leave-One-Out (LOOCV)

$k = n$. Uses all but one observation for training.

- ▶ Pro: Nearly unbiased.
- ▶ Con: Computationally expensive.

Repeated CV

Repeats the k -fold process multiple times with different splits to reduce variance in the estimate.

The Bias-Variance Trade-off

Bias (Underfitting)

Error from erroneous assumptions (e.g., assuming data is linear when it's quadratic).

- ▶ Misses relevant relations.
- ▶ **Fix:** Increase complexity, add features.

Variance (Overfitting)

Error from sensitivity to small fluctuations in the training set.

- ▶ Models the random noise.
- ▶ **Fix:** Regularization, more data, simplify model.

Goal: Find the sweet spot where Total Error ($\text{Bias}^2 + \text{Variance}$) is minimized.

Summary and Key Takeaways

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- ▶ **Selection vs Evaluation:** Don't use your test set to tune parameters!
 - ▶ **Metrics:** Accuracy is not enough. Use Precision/Recall/AUC for a complete picture.
 - ▶ **Information Criteria:** AIC and BIC allow model comparison without a hold-out set, penalizing complexity.
 - ▶ **Robustness:** Cross-Validation (especially Stratified) is the gold standard for performance estimation.
 - ▶ **The Trade-off:** Machine Learning is largely the art of balancing Bias and Variance.